

Abstract No.

RAD52 Proteomic Analysis in Glioblastoma Cell Hierarchy

Candela Arribas-Parejas¹, Alejandro Pérez Recio Pérez-Recio¹, and Diana Aguilar-Morante^{1,}*

¹Institute of Biomedicine of Seville, IBiS/Virgen del Rocío University Hospital/CSIC/University of Seville, Spain

*Presenting author

Glioblastoma is the most aggressive primary brain tumour in adults and is characterized by profound cellular heterogeneity and inevitable recurrence following standard therapy. This recurrence is driven by Glioblastoma Stem-like Cells (GSCs), a subpopulation with high self-renewal capacity, enhanced DNA repair activity, and resistance to cytotoxic stress. Defining the molecular programs that sustain this hierarchy and uncovering vulnerabilities in DNA repair pathways remain central challenges in glioblastoma biology. Here, we used comparative proteomics to map functional differences between GSCs and Differentiated Glioblastoma Cells (DGCs). The analysis revealed a clear dichotomy in cellular states, with GSCs exhibiting a proliferative, biosynthetically active, and DNA repair-associated phenotype, while DGCs showed signatures consistent with metabolic maturation and increased extracellular matrix interaction. We next investigated the impact of RAD52 depletion, a key mediator of homologous recombination and replication stress tolerance. RAD52 inhibition triggered modest but coherent proteomic shifts indicative of increased cellular stress and reduced mesenchymal-associated programs. Despite these changes, core stem-like signatures remained largely stable, suggesting that RAD52 is not essential for maintaining stem-like identity under the conditions tested. Together, these findings define distinct proteomic programs underlying glioblastoma cellular hierarchy and suggest that RAD52 primarily supports stress adaptation and phenotypic plasticity rather than stemness maintenance. These results highlight DNA repair dependencies as potential therapeutic entry points to impair tumour adaptability without directly eroding the stem-like compartment.



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